

Cumulative frequency tables



1

a 26

b 6

c 18

2

Score (x)	Frequency (f)	Cumulative frequency (cf)
12	8	8
13	5	13
14	8	21
15	5	26

3

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
1 → 4	4	4
5 → 8	5	9
9 → 12	9	18
13 → 16	5	23
17 → 20	4	27

b 27

c 4

4

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
1 → 5	15	15
6 → 10	26	41
11 → 15	18	59
16 → 20	14	73
21 → 25	7	80
26 → 30	2	82

b 82

c 5

d 10

5



a

Score (x)	Frequency (f)	Cumulative frequency (cf)
20 → 24	7	7
25 → 29	18	25
30 → 34	25	50
35 → 39	12	62
40 → 44	8	70
45 → 49	4	74
50 → 54	1	75

b 75

c 5

d 34

Cumulative frequency graphs

6

Score	Frequency	Cumulative frequency
25	10	10
26	16	26
27	14	40
28	20	60
29	16	76
30	10	86

7

Score	Frequency
78	3
79	3
80	5
81	4
82	3

8

a 14

b 2

c 0

d 71.4%

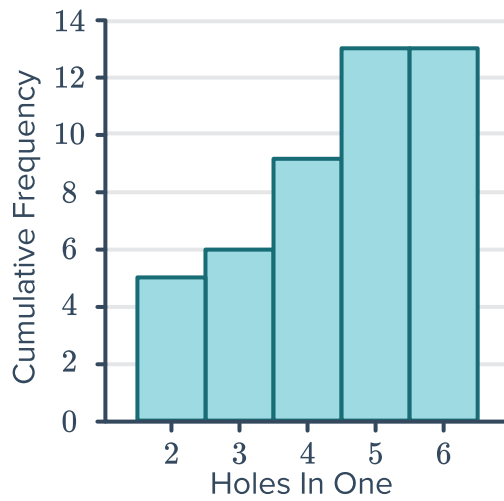
9

a



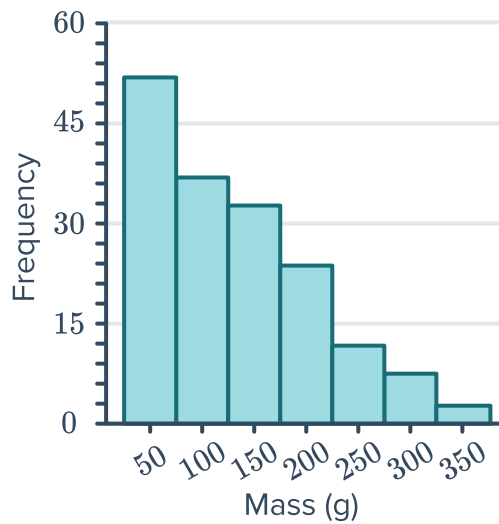
b

45



c 6

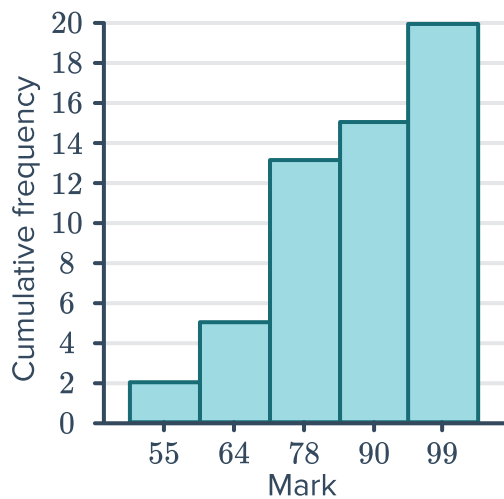
10



11

a

b 15 students



c 5 students

d 2 students



Applications

12

a

Grade	Score (x)	Frequency (f)	Cumulative frequency (cf)
E	$0 \leq x < 20$	7	7
D	$20 \leq x < 40$	14	21
C	$40 \leq x < 60$	32	53
B	$60 \leq x < 80$	97	150
A	$80 \leq x < 100$	62	212

b 212

c 20

d 60

13

a 2

b 12

c

d 7

Score (x)	Frequency (f)	Cumulative frequency (cf)
2	1	1
3	2	3
4	5	8
5	5	13
6	5	18
7	9	27
8	7	34
9	5	39
10	8	47
11	1	48
12	2	50

e 11

f 18

14



a

Number of sightings	Number of locations (f)	Cumulative frequency (cf)
7	1	1
8	4	5
9	5	10
10	2	12
11	1	13
12	7	20

b 19 locations

c 12 locations

15

a

Time recorded	Frequency	Cumulative frequency
$16:00 \leq x < 16:10$	4	4
$16:10 \leq x < 16:20$	5	9
$16:20 \leq x < 16:30$	3	12
$16:30 \leq x < 16:40$	4	16
$16:40 \leq x < 16:50$	2	18
$16:50 \leq x < 17:00$	7	25

b 25

c 16

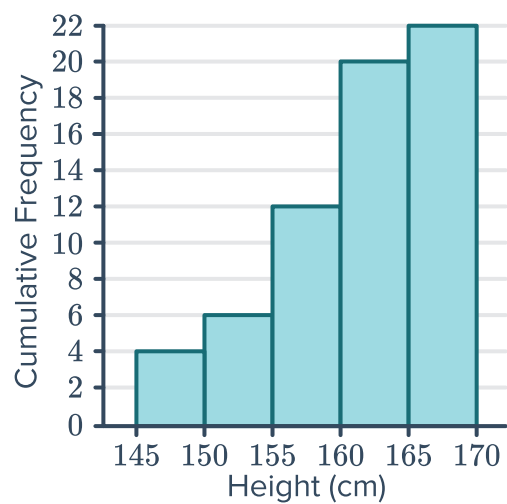
d 48%

16

a



b 16 students



c 6 students

d 2 students

e 153

17

a The difference between the heights of two adjacent columns.

b 7